

An Attack-Free Safety Monitor for Catastrophic Overfitting in Fast Adversarial Training

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Abstract—Fast adversarial training is efficient but can abruptly lose multi-step robustness through catastrophic overfitting (CO). A standard randomized FGSM recipe is often stable, so a useful safeguard should stay inactive in that case and escalate only when training becomes unsafe. We propose an attack-free online monitor based on the drop in cosine similarity between two input gradients already produced by an FGSM parameter update. A threshold selected on three development seeds is frozen before all subsequent tests. Across seven held-out CIFAR-10 failure events, the monitor warns within 60 optimizer updates in 7/7 cases, with a median lead of 40 updates and 0.143 early false alarms per run; six events become terminal CO and one is transient. On three stable FGSM-RS runs it produces no alarms. On the identical traces, a fixed-cadence PGD-10 monitor detects 6/7 events with a median delay of 40 updates after making a median of 73 PGD calls before failure. Triggering a switch to two-step PGD raises final validation PGD-10 accuracy in a deliberately collapse-prone regime from $0.14 \pm 0.19\%$ to $46.72 \pm 1.05\%$ across five seeds while retaining $83.09 \pm 0.57\%$ clean accuracy. A matched rollback ablation changes PGD-10 by only +0.11 percentage points on average, is beneficial on 3/5 seeds, and adds 0.53 minutes, so rollback is not part of the simplified policy. Strong evaluation gives 44.75% full-test PGD-50 accuracy and 41.30% AutoAttack accuracy on 1,000 examples. The method is therefore a conditional safety monitor, not a replacement for a well-tuned FGSM-RS recipe or full PGD training.

I. INTRODUCTION

Deep classifiers can change their predictions under small, deliberately chosen input perturbations [1], [2]. This failure is particularly concerning when a model is deployed in a safety- or security-sensitive pipeline. Adversarial training addresses the problem by optimizing against perturbed examples and remains one of the strongest empirical defenses [3]. Its main obstacle is cost: a multi-step inner maximization is solved for nearly every parameter update.

Fast adversarial training replaces this inner loop with a single fast gradient sign method (FGSM) step. With an appropriate initialization and schedule, it can approach the robustness of PGD training at a fraction of the computation [4]. However, the same procedure can exhibit *catastrophic overfitting* (CO): accuracy against iterative attacks drops from a useful value to almost zero over a short part of training, even when accuracy against the training attack remains high. Monitoring the model with a separate PGD attack can identify the failure, but repeated validation attacks partly offset the efficiency that motivates fast training.

This paper investigates a narrower question: can the geometry already exposed by an FGSM update forecast CO

early enough to support conditional escalation? An FGSM update computes an input gradient at the attack starting point, evaluates the network at the perturbed endpoint, and backpropagates the endpoint loss to update model parameters. By retaining the endpoint input gradient, the cosine similarity of the two input gradients is available without another forward or backward pass. We use a local drop in this cosine as a warning score. If it crosses a frozen threshold, the current FGSM epoch finishes and subsequent epochs use two-step PGD (PGD-2). Stable training is left unchanged. This positioning matters: FGSM-RS is already a strong and faster reference in our experiments, whereas the monitor is intended to expose unsafe trajectories caused by recipe or configuration changes.

The contribution is threefold:

- We formulate an attack-free monitor based on an adaptive drop of within-update gradient consistency. After selection on three development seeds, its threshold is frozen and evaluated on seven held-out failure events plus three stable FGSM-RS runs. A direct replay against fixed-cadence PGD-10 monitoring separates forecasting lead from attack-based detection.
- We separate detection from mitigation through matched controls: detector-guided PGD-2 with and without rollback, fixed switches at epochs 13 and 21, and PGD-2 from the first epoch. The results remove rollback from the final policy rather than attributing its effect to the detector.
- We evaluate forecasting, clean and robust accuracy, and measured training time separately, then test frozen checkpoints with PGD-10, PGD-20, PGD-50, five-restart PGD, and the official AutoAttack implementation [5].

The proposed method is an empirical monitor and training safeguard. It neither claims a formal robustness certificate nor claims higher robustness than every fixed fast-training recipe. Certified defenses bound worst-case behavior by other means and are complementary to the present objective.

II. RELATED WORK

A. Adversarial Attacks and Training

FGSM follows the sign of the input loss gradient in one step [2]. PGD iterates gradient steps with projection onto a prescribed threat set and is central to the robust-optimization view of adversarial training [3]. TRADES separates natural error from a boundary-related robustness term and makes the clean-robust trade-off explicit [6]. Although these methods

are effective, their multi-step inner loops increase training cost. Free adversarial training reuses gradients across repeated mini-batches [7], whereas fast training uses a single randomized FGSM step [4].

Robustness evaluation itself must be sufficiently strong. PGD results can be sensitive to iterations, restarts, and objectives. AutoAttack combines complementary parameter-free attacks to reduce evaluator bias and expose gradient masking [5]. We therefore report progressively stronger PGD variants and AutoAttack rather than treating FGSM accuracy as a robustness estimate.

B. Catastrophic Overfitting

Wong *et al.* identified CO as a failure mode of fast adversarial training [4]. Andriushchenko and Flammarion connected it to local nonlinearity and proposed GradAlign, which regularizes gradient alignment inside the perturbation set [8]. Li *et al.* showed that recovery from weak-attack overfitting is important and proposed FastAdv+ and FastAdvW, which use periodic PGD validation to decide when to introduce PGD updates [9]. Our response mechanism is related in spirit, but the detector differs operationally: it consumes the two input gradients already present in the ordinary FGSM parameter update and does not execute a validation attack. We also separate prediction from mitigation by freezing the detector before held-out evaluation.

C. Empirical and Certified Robustness

Empirical defenses are evaluated against known attacks but do not guarantee correctness for every allowed perturbation. Certified methods instead compute a provable lower bound, for example through convex outer approximations [10] or randomized smoothing under an ℓ_2 threat model [11]. This paper targets efficient empirical training in the ℓ_∞ setting. The distinction is important: the attack results reported below measure resistance to a strong test suite, not certified radii.

III. METHOD

A. Threat Model and Fast Update

Let f_θ be a classifier, ℓ the cross-entropy loss, and (x, y) a training example. Under the untargeted ℓ_∞ threat model, adversarial training approximates

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_\infty \leq \epsilon} \ell(f_\theta(x + \delta), y) \right]. \quad (1)$$

FGSM constructs one endpoint

$$\begin{aligned} g^0 &= \nabla_{x^0} \ell(f_\theta(x^0), y), \\ x^{\text{adv}} &= \Pi_{[0,1] \cap B_\infty(x,\epsilon)} (x^0 + \alpha \text{sign}(g^0)). \end{aligned} \quad (2)$$

where x^0 is either the clean input or a random point in the perturbation set. The endpoint loss is then backpropagated for the parameter update.

B. Free Gradient-Consistency Signal

The endpoint backward pass also gives the retained gradient $g^1 = \nabla_{x^{\text{adv}}} \ell(f_\theta(x^{\text{adv}}), y)$. For each example, the local consistency is

$$c(x) = \frac{\langle g^0, g^1 \rangle}{\|g^0\|_2 \|g^1\|_2}. \quad (3)$$

The batch values are averaged over non-overlapping windows of 20 optimizer updates, producing c_t . This statistic adds reductions and storage but no model evaluation: g^0 is required to construct FGSM, while g^1 is produced by the same backward pass that calculates the parameter gradient.

Absolute cosine values vary through training. We therefore compare the current window with its recent history. Let

$$b_t = \text{median}(c_{t-5}, \dots, c_{t-1}), \quad s_t = b_t - c_t, \quad (4)$$

using all available history after at least three prior windows. The detector fires when $s_t \geq \tau$. Development runs with seeds 17, 23, and 42 determine $\tau = 0.1663298875$; the window, history rule, and threshold are then frozen.

C. Conditional PGD-2 Escalation

The simplified policy performs three operations when a warning occurs:

- 1) record the first warning time;
- 2) finish the current epoch with the original FGSM update; and
- 3) train all subsequent epochs with PGD-2 using step size $4/255$.

PGD-2 is a moderate-cost recovery adversary. The design is asymmetric: FGSM is used while its local geometry is stable, and computation increases only after evidence of instability. An initially proposed rollback variant snapshots and restores the model, optimizer, mixed-precision scaler, data-loader state, and random states at the start of an affected epoch before replaying it with PGD-2. Because the matched ablation in Section IV does not show a consistent benefit, rollback is retained only as an ablation and not as part of the final policy.

D. Forecasting Protocol

A collapse event is defined before detector evaluation. On a fixed validation batch, PGD-10 accuracy must fall by at least 20 percentage points from its previous peak and reach at most 10% within one 20-update window. Windows in the three-window interval preceding the first event are positive forecast targets. We report event recall, false alarms per run, median warning lead, AUROC, and AUPRC. The threshold and event definition are selected using seeds 17, 23, and 42. They remain frozen for held-out seeds 101, 202, 303, 404, 505, 606, and 707. For the five new held-out runs, a fixed validation batch is labeled by PGD-10 every five trace windows and for three windows after a warning. These attacks are used only for retrospective event labels and never affect training or the detector. We additionally apply the frozen detector in observation-only mode to three stable FGSM-RS runs; every warning in a non-collapsing run counts as a false alarm.

TABLE I
FROZEN DETECTOR ON SEVEN HELD-OUT FAILURE EVENTS

Seed	Event epoch	Lead	Early FA	Outcome
101	21	60	0	terminal
202	18	40	1	terminal
303	13	40	0	transient
404	22	20	0	terminal
505	20	52	0	terminal
606	23	40	0	terminal
707	24	60	0	terminal

IV. EXPERIMENTS

A. Setup

We use CIFAR-10 [12] and PreActResNet-18 [13]. A deterministic 45,000/5,000 training/validation split is used for development and checkpoint selection. The threat radius is $\varepsilon = 8/255$. All methods use SGD with momentum 0.9, weight decay 5×10^{-4} , batch size 128, mixed precision, and the same GPU.

Collapse and intervention runs last 30 epochs with a triangular learning-rate schedule peaking at 0.3. Before escalation, FGSM starts at the clean input and uses step size $8/255$; this deliberately challenging recipe exposes CO. Escalation uses PGD-2 with step size $4/255$. Fixed-switch controls use exactly the same code and tracing, but ignore detector warnings and switch only at the specified epoch. The stable Fast FGSM-RS reference uses random initialization and a maximum learning rate of 0.2. The reference PGD-10 model is trained for 110 epochs with step size $2/255$.

Final checkpoints are selected by validation PGD-10 accuracy. Clean, FGSM, PGD-10, PGD-20, PGD-50, and five-restart PGD-20 are evaluated on all 10,000 test examples. Standard AutoAttack (APGD-CE, targeted APGD, targeted FAB, and Square Attack) uses the official implementation and the same fixed first 1,000 test examples for every model. Thus AutoAttack values are comparable within Table IV, but are not full-test estimates.

B. Frozen-Detector Evaluation

Figure 1 illustrates the forecasting mechanism on five unprotected traces. PGD-10 accuracy deteriorates near the registered event, whereas the cosine-drop score rises beforehand. This temporal ordering is the basis for using the statistic as a warning rather than a post-collapse test.

The frozen detector is then applied to seven held-out seeds (101, 202, 303, 404, 505, 606, and 707). Table I reports every event, rather than only an aggregate percentage. All seven events are warned within the predeclared 60-update horizon. Lead time ranges from 20 to 60 updates, with a median of 40. There is one early false-alarm episode in total (0.143 per run). Six trajectories end below 10% PGD-10; seed 303 is a transient event that subsequently recovers, and is identified separately to avoid equating an event warning with terminal failure.

We also test the detector in observation-only mode on stable FGSM-RS seeds 17, 23, and 42. None collapses and

TABLE II
DIRECT ONLINE-MONITORING BASELINE ON IDENTICAL TRACES

Monitor	Event recall	Median timing	PGD calls
Periodic PGD-10/100 updates	6/7	40 after	73 before event
Frozen gradient-drop (ours)	7/7	40 before	0

TABLE III
MATCHED THREE-SEED COMPONENT CONTROLS (MEAN $\pm$ SAMPLE STD.)

Policy	Clean	PGD-10	Core min
Detector + rollback	82.95 $\pm$ 0.55	46.88 $\pm$ 0.51	18.83 $\pm$ 0.94
Detector, no rollback	82.92 $\pm$ 0.49	47.16 $\pm$ 0.12	18.31 $\pm$ 0.75
Fixed switch, epoch 13	83.19 $\pm$ 0.69	46.51 $\pm$ 0.25	19.23 $\pm$ 0.05
Fixed switch, epoch 21	81.97 $\pm$ 1.23	45.83 $\pm$ 1.06	17.70 $\pm$ 0.02
PGD-2 from epoch 1	83.40 $\pm$ 1.02	47.21 $\pm$ 0.47	21.58 $\pm$ 0.18

the detector emits zero alarms across all three runs. Their final validation PGD-10 accuracies are 46.25%, 47.81%, and 46.80% (mean 46.95%). Hence the frozen rule is inactive on the stable recipe tested here; no training-time attack is used either to trigger or suppress it.

As a direct attack-based monitoring baseline, we replay the same saved trajectories with one PGD-10 validation batch every five trace windows (100 updates). The baseline applies the same 20-point/10% event rule and ignores the extra labels collected after our warnings. This is a controlled analogue of periodic-PGD monitoring used by recovery methods such as FastAdv+ [9], rather than a reimplement of its complete training policy. Table II shows the operational difference. The periodic monitor detects all six terminal events but misses the transient seed 303 event between probes. Its median detection delay is 40 updates, whereas the proposed signal has 40 updates of median lead. Before an event, periodic monitoring has already invoked PGD-10 a median of 73 times per run; the proposed online decision invokes it zero times.

C. Mitigation and Matched Component Controls

Across five seeds, unprotected FGSM retains $83.27 \pm 1.77\%$ clean accuracy but finishes at only $0.14 \pm 0.19\%$ validation PGD-10 accuracy. The simplified detector-guided policy without rollback raises these values to $83.09 \pm 0.57\%$ clean and $46.72 \pm 1.05\%$ PGD-10, with core training time 18.14 ± 0.58 min versus 15.53 ± 0.16 min. Thus clean accuracy alone would not reveal the failure, while conditional escalation prevents terminal CO in all five intervention runs.

Table III holds seeds fixed at 17, 101, and 202 to separate the detector, rollback, and PGD-2. The no-rollback policy is slightly stronger than rollback in this matched subset. Over the complete five paired seeds, rollback changes PGD-10 by only +0.11 percentage points on average, wins on 3/5 seeds, and adds 0.53 min; it is therefore excluded from the final method.

The detector switches at epochs 17, 21, and 13 for these three seeds. An early fixed switch spends more PGD-2 computation yet is 0.65 points weaker than the detector policy. A late epoch-21 switch permits transient near-zero

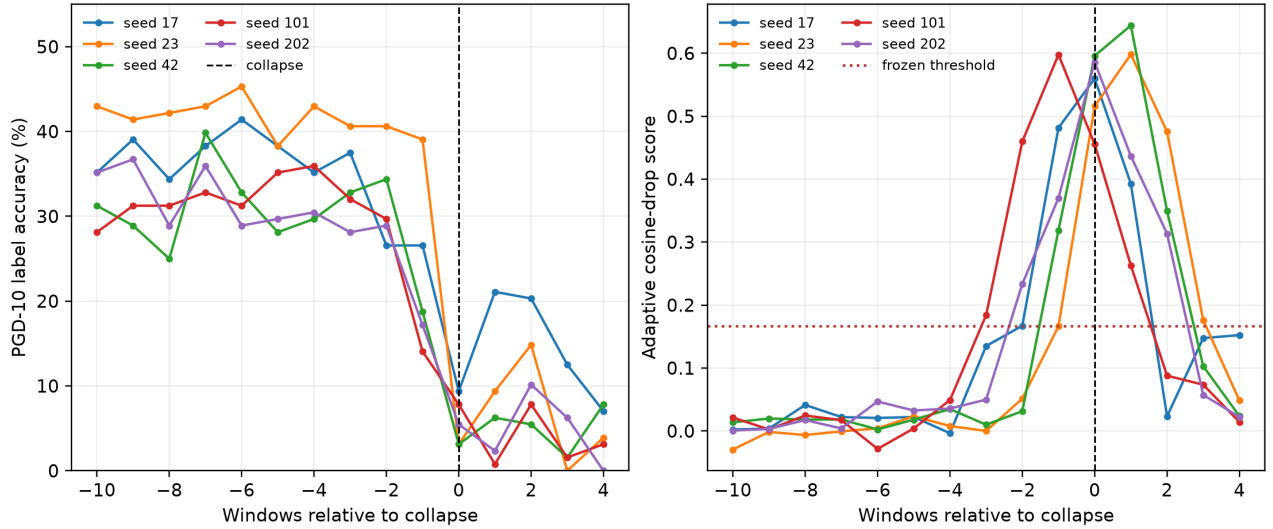


Fig. 1. FGSM traces aligned at the first registered collapse. Left: validation PGD-10 accuracy. Right: adaptive cosine-drop score; the dotted line is the threshold frozen on three development seeds. One window is 20 optimizer updates.

TABLE IV
UNIFIED CIFAR-10 TEST ACCURACY (%). AUTOATTACK USES A
FIXED 1,000-IMAGE SUBSET; OTHER COLUMNS USE ALL 10,000
IMAGES.

Method	Clean	PGD-10	PGD-50	PGD×5	AA
Detector, no rollback	82.57	46.38	44.75	44.91	41.30
Detector + rollback	82.97	47.20	45.46	45.65	42.60
Fast FGSM-RS	82.48	47.77	46.24	46.36	43.00
PGD-10 training	82.47	51.76	50.77	50.84	48.70

PGD-10 accuracy in two runs and gives the weakest final result. Full PGD-2 is only 0.05 points stronger than the detector policy but costs 3.27 additional minutes on average. These controls do not imply universal superiority over fixed recipes; they show that the warning provides useful timing information within the collapse-prone regime.

D. Strong Test-Set Evaluation

Table IV uses one evaluator for frozen checkpoints. The final no-rollback policy retains 44.75% accuracy under PGD-50 and 44.91% under five-restart PGD-20. Their small gap indicates convergence of the tested PGD configurations. AutoAttack gives the more conservative 41.30% estimate.

Fast FGSM-RS is both stronger and faster than the simplified monitored model in this experiment: its AutoAttack accuracy is 1.70 points higher and its representative core time is 16.04 min versus 17.50 min. PGD-10 training is strongest but takes 262.48 min. These results determine the scope of the contribution: the monitor is not a new robustness optimum or a reason to avoid a well-tuned FGSM-RS recipe. It is an attack-free online safety mechanism that stays silent on the stable recipe and conditionally protects fast-training configurations when their local geometry becomes unsafe.

E. Discussion

The evidence supports two bounded conclusions. First, a frozen local gradient-consistency rule forecasts all seven observed held-out gradient events and does not fire in three stable control runs. Second, the matched intervention study attributes recovery to PGD-2 escalation, not rollback; the detector’s role is to choose when to pay that cost. The agreement among iterative PGD attacks and AutoAttack also rules out conclusions based solely on the weak training attack. Certification remains complementary: the present experiments measure empirical robustness and do not establish a formal guarantee.

V. COMPARISON, COST, AND TRANSFER SCOPE

A. Direct Baselines under One Evaluator

The strongest comparison is the one that holds architecture, data split, threat radius, checkpoint rule, and evaluator fixed. Table IV does so for the proposed policy, Fast FGSM-RS, and PGD-10 training. Relative to Fast FGSM-RS, the no-rollback policy is 1.39 points lower under PGD-10, 1.49 points lower under PGD-50, and 1.70 points lower under AutoAttack. It also takes 9.1% more core training time (17.50 versus 16.04 min). These numbers rule out a claim that monitoring improves an already well-tuned FGSM-RS run. Relative to PGD-10 training, the policy is 7.40 points lower under AutoAttack, but PGD-10 training is approximately 15 times slower (262.48 versus 17.50 min). The useful comparison is therefore not “best robustness at any cost,” but whether an unsafe fast-training trajectory can be detected without adding a second validation attack and stabilized with conditional computation.

Within that target regime, the paired comparison is decisive: unprotected zero-start FGSM ends at $0.14 \pm 0.19\%$ validation PGD-10, whereas monitored training ends at $46.72 \pm 1.05\%$. The stable FGSM-RS control establishes the other required behavior: when the fast recipe is already

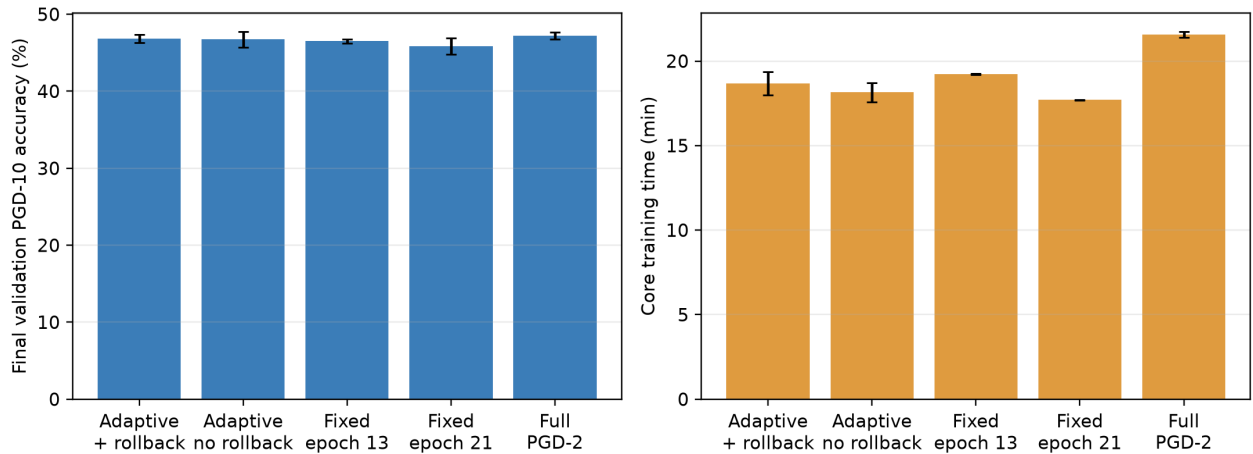


Fig. 2. Component controls. Detector variants use five seeds; fixed schedules and full PGD-2 use the matched three-seed subset. Error bars are sample standard deviations. The table provides the strictly matched comparison.

safe, the detector produces no alarms in three runs and therefore requests no PGD-2 escalation. Together, these controls distinguish a safety mechanism from a generally stronger adversarial-training algorithm.

B. Relationship to CO Countermeasures

Table V compares the operating assumptions of the closest families. Fast FGSM-RS combines randomized initialization with a cyclic schedule and commonly uses PGD-based early stopping [4]. FastAdv+ and FastAdvW explicitly monitor robustness with periodic PGD validation and introduce PGD updates when weak-attack overfitting is observed [9]. GradAlign instead penalizes disagreement between input gradients at different points in the perturbation set, directly regularizing the local nonlinearity associated with CO [8]. Our method shares GradAlign’s geometric motivation and FastAdv+’s conditional response, but has a different systems contract: its online evidence comes from gradients already generated by the ordinary FGSM update, and the signal is frozen and assessed separately from the PGD-2 response.

This table is a mechanism-level comparison rather than a cross-paper accuracy ranking. Published numbers use different networks, schedules, attack settings, and hardware, so copying them beside our measurements would create false precision. The quantitative baselines in Table IV are rerun under our evaluator; Table V states what is and is not shared with earlier CO countermeasures. A fully controlled numerical comparison with the official FastAdv+ and GradAlign implementations remains distinct from the evidence presented here.

C. What Generalizes in the Present Evidence?

The detector is not evaluated only on the two original collapse traces. Its threshold is chosen once on seeds 17, 23, and 42, then held fixed across seven new failure seeds whose event epochs range from 13 to 24. This tests transfer across optimization noise, substantially different failure times, a transient event, and six terminal collapses. The same frozen

rule is also transferred without adjustment to three stable FGSM-RS runs, where it remains silent. Thus the current evidence supports seed-level and trajectory-level transfer within CIFAR-10/PreActResNet-18. It does not by itself establish transfer to a new dataset or architecture. We keep this boundary explicit because retuning the threshold on a second setting and calling the result generalization would weaken the central frozen-detector claim.

D. Reproducibility and Audit Trail

All detector choices are made before held-out evaluation: 20-update windows, a five-window median history, at least three history windows, threshold $\tau = 0.1663298875$, a 60-update forecast horizon, and the event definition in Section III-D. The seven held-out event rows are reported individually, and the stable controls count every warning as a false alarm. Fixed-switch ablations explicitly disable detector-triggered switching, preventing the nominal fixed schedules from silently becoming adaptive. Training time sums only core update time; validation and retrospective label attacks are excluded for all methods. Finally, all frozen checkpoints are evaluated with the same full-test PGD implementations, while the smaller AutoAttack scope is marked in the table rather than mixed with the 10,000-example estimates. These choices make the detector, intervention, and evaluator independently auditable.

VI. CONCLUSION

This work presented an attack-free safety monitor for catastrophic overfitting in fast adversarial training. The input gradient used to construct FGSM and the endpoint gradient created by the ordinary parameter backward pass expose a predictive local-geometry signal without a validation attack. With its threshold frozen after three development seeds, the monitor warns before all seven held-out failure events and remains silent in three stable FGSM-RS runs. On the same traces, fixed-cadence PGD-10 monitoring detects only 6/7 events and does so after rather than before the event,

TABLE V
OPERATIONAL COMPARISON WITH CLOSELY RELATED FAST-AT STRATEGIES

Method	Primary purpose	Online evidence	Extra online computation	Action on instability
Fast FGSM-RS [4]	Stable fast training	PGD checkpoint/early-stop test	Validation attack	Stop/select checkpoint
FastAdv+/W [9] GradAlign [8]	Detect and recover Prevent CO	Periodic PGD robustness Random-point gradient alignment	Validation attack Extra gradient regularizer	Temporary/permanent PGD Modify every FGSM update
Ours	Forecast and safeguard	Within-update gradient drop	Reductions and trace state	Conditional PGD-2 switch

while requiring repeated validation attacks. Matched interventions show that switching to PGD-2 prevents terminal collapse, whereas rollback contributes no consistent average improvement and is removed from the final policy. The resulting model reaches 44.75% PGD-50 accuracy on the full CIFAR-10 test set and 41.30% standard AutoAttack accuracy on a fixed 1,000-image subset. These results support the monitor as a low-cost conditional safeguard, not as a claim to supersede the stable FGSM-RS reference or full PGD training. Evaluation on additional architectures and datasets, and the integration of empirical monitoring with certified defenses, remain important next steps.

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